

REDFLAG

Transaction Anomaly Detection Using SQL

1. Introduction

RedFlag is a transaction analysis project developed using SQL and MySQL. The purpose of this project is to analyze transaction data and identify unusual transaction patterns that may require further investigation.

The project analyzes information such as transaction amount, transaction status, user ID, merchant ID, transaction time, payment mode, city, and transaction type.

SQL queries are used to find failed transactions, high-value transactions, repeated transactions, users with unusually high transaction activity, and other suspicious patterns.

2. Problem Statement

Large transaction datasets can contain unusual or suspicious patterns that are difficult to identify manually.

The problem is to analyze transaction data using SQL and identify patterns such as:

- High numbers of failed transactions
- Repeated transactions with the same amount
- Very high transaction frequency
- High-value transactions
- Users showing unusual transaction behavior

These patterns can then be treated as Red Flags for further investigation.

3. Objectives

The main objectives of the RedFlag project are:

1. To analyze a large transaction dataset using SQL.
2. To identify successful and failed transactions.
3. To identify high-value transactions.
4. To find users with a high number of failed transactions.
5. To calculate user failure rates.

6. To identify repeated transactions.
7. To detect unusually high transaction frequency.
8. To identify suspicious or unusual transaction patterns.
9. To summarize the findings for further investigation.

4. Tools and Technologies

- **Database:** MySQL
- **Interface:** MySQL Workbench
- **Language:** SQL
- **Dataset:** Transaction dataset
- **Analysis:** SQL queries and aggregation functions

5. Dataset Description

The dataset contains **200,594 transactions**.

The important columns are:

Column	Description
TXN ID	Unique transaction ID
User ID	Identifies the user
Merchant ID	Identifies the merchant
Amount	Transaction amount
TXN Time	Date and time of transaction
Status	SUCCESS or FAILED
Payment Mode	Method used for payment
City	City of transaction
TXN Type	Type of transaction

6. Overall Transaction Analysis

The analysis produced the following results:

Metric	Result
Total Transactions	200,594
Successful Transactions	194,272
Failed Transactions	6,322
Average Transaction Amount	₹3,932.22
Highest Transaction Amount	₹99,882.92
Transactions Above ₹90,000	49
Failed Transactions Above ₹90,000	0

The dataset contains a large number of successful transactions, while a smaller portion of transactions failed.

7. Failed Transaction Analysis

SQL was used to identify users with a high number of failed transactions.

There were **25 users with 20 or more failed transactions**.

The top users included:

User ID	Total Transactions	Failed Transactions	Failure Rate
14595	70	35	50%
14593	68	34	50%
14576	66	33	50%
14580	64	32	50%
14581	64	32	50%
14599	64	32	50%
14585	62	31	50%
14589	62	31	50%
14590	62	31	50%

User ID Total Transactions Failed Transactions Failure Rate

14591 62 31 50%

These users show an unusually high failure rate of **50%**, making them important candidates for further investigation.

8. Repeated Transaction Analysis

The project also searched for users making the same transaction amount repeatedly.

The most repeated amount was **₹9,999**.

User ID Amount Repeat Count

14690 ₹9,999 25

14680 ₹9,999 25

14693 ₹9,999 22

14689 ₹9,999 21

14684 ₹9,999 21

14691 ₹9,999 20

14695 ₹9,999 17

14686 ₹9,999 17

14679 ₹9,999 16

14678 ₹9,999 15

User **14690** made the same ₹9,999 transaction **25 times**.

This repeated pattern is considered a Red Flag that may require further investigation.

9. High-Frequency Transaction Analysis

The project also analyzed how many transactions a user made on the same day.

The highest observed activity was:

- **User ID:** 14569
- **Date:** 03-April-2024

- **Transactions:** 60
- **Total Amount:** ₹323.58
- **Average Amount:** ₹5.39
- **Highest Transaction:** ₹9.96
- **Status:** All 60 were successful

Making 60 transactions in one day is an unusual frequency pattern and therefore can be considered a Red Flag.

10. Suspicious User Analysis

Using the project criteria:

- 20 or more failed transactions, **OR**
- 60 or more total transactions

the analysis identified **90 users** matching at least one of these conditions.

These users should not automatically be considered fraudulent. Their activity should instead be reviewed using additional information and business rules.

11. Payment Mode Analysis

For the three highest failed-transaction users:

User IDs: 14595, 14593 and 14576

The analysis showed:

- Payment Mode: **Card**
- Total transactions: **204**
- Failed transactions: **102**
- Failure rate: **50%**

This indicates that card transactions for these users had a high failure rate.

12. City Analysis

For the same three users:

City	Transactions	Failed
Kolkata	70	35
Delhi	68	34
Mumbai	66	33

Each of these users showed a 50% failure rate in the analyzed data.

13. Transaction Type Analysis

For the same three users:

- Transaction Type: **Debit**
- Total transactions: **204**
- Failed transactions: **102**
- Failure rate: **50%**

14. Key Red Flags Identified

The major Red Flags found in the project are:

Red Flag 1 — High Failure Rate

Several users had a 50% transaction failure rate.

Red Flag 2 — Repeated Transactions

Multiple users repeatedly made ₹9,999 transactions.

Red Flag 3 — High Transaction Frequency

One user made 60 transactions on a single day.

Red Flag 4 — High-Value Transactions

49 transactions were above ₹90,000. However, none of these high-value transactions were failed.

15. Conclusion

The RedFlag project demonstrates how SQL can be used to analyze a large transaction dataset and identify unusual transaction behavior.

The analysis successfully identified users with high failure rates, repeated transaction amounts, and unusually high transaction frequency.

These patterns can be used as warning signals in a transaction monitoring system. However, a Red Flag does not prove that a transaction or user is fraudulent. Additional investigation and business rules would be required before making such a decision.

16. Future Scope

The project can be improved in the future by:

1. Adding real-time transaction monitoring.
 2. Using Python for advanced data analysis.
 3. Creating dashboards using Power BI or Tableau.
 4. Adding machine-learning-based fraud detection.
 5. Sending automatic alerts for suspicious transactions.
 6. Adding location and time-based anomaly detection.
 7. Building a real-time fraud detection system.
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17. Final Project Summary

Project Name: RedFlag – Transaction Anomaly Detection Using SQL

Technology: MySQL and SQL

Dataset Size: 200,594 transactions

Main Purpose: Identify unusual transaction patterns

Major findings:

- 6,322 failed transactions
- 25 users with 20+ failed transactions
- 90 users matching the final suspicious-user criteria
- ₹9,999 was the most repeated transaction amount
- Maximum repeated ₹9,999 transactions by one user: 25
- Maximum same-day transactions by one user: 60
- Highest transaction amount: ₹99,882.92